BLOCKCHAIN INTELLIGENCE · CRYPTO TRACING EXERCISE

LockFlow Ransomware — Fund Flow Analysis

Open-source transaction graph investigation using Breadcrumbs.app

■ This is a self-directed learning exercise. I used the publicly available Breadcrumbs.app tool to reconstruct the fund flow of a fictitious ransomware case. Wallet addresses and amounts are invented, but the methodology and analytical reasoning follow real-world crypto tracing practice.

Reference	**CTR-2024-0018**	Date	**14 June 2024**
Analyst	**J.M. Cánovas Soler**	Tool	**Breadcrumbs.app (free tier)**
Network	**Bitcoin (BTC)**	Status	**Learning exercise — not a live case**

! **Main finding: ransomware proceeds — funds partially laundered**
BTC traced victim → ransom wallet → mixer → exchange. ~68% reached a fiat off-ramp. Remaining 1.01 BTC could not be traced beyond the mixer output.

1. CASE BACKGROUND

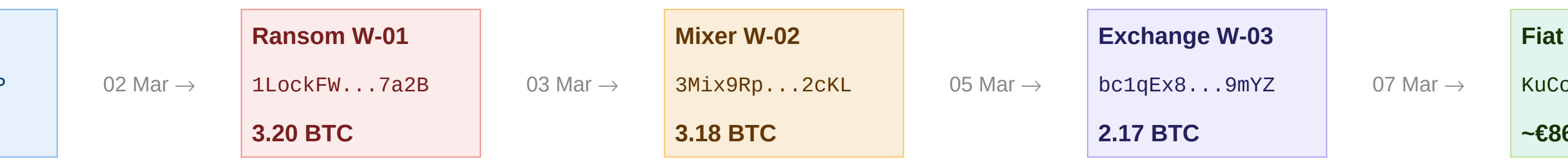
Incident type	Ransomware extortion — LockFlow group
Victim entity	LogiFreight GmbH (fictitious — DE)
Ransom demanded	3.2 BTC (~€128,000 at time of payment)
Payment date	02 March 2024 — 14:32 UTC
Initial receiving wallet	**1LockFW9xK3Rp...7a2B (W-01)**
Investigation scope	Full transaction graph — W-01 to fiat off-ramp

■ I chose a ransomware scenario because it's one of the most common crypto crime typologies in current AML/financial crime job descriptions. The LockFlow name is fictitious but the fund movement pattern is based on publicly documented ransomware cases (e.g. Colonial Pipeline).

2. TRANSACTION GRAPH — FUND FLOW

The diagram below shows the reconstructed transaction chain, built manually in Breadcrumbs.app by following each outgoing transaction from the initial ransom wallet.

Victim payment | Ransom wallet | Mixer | Exchange | Fiat exit



Mixer confirmed | Exchange deposit identified | 1.01 BTC unaccounted | 68% traced to fiat

■ Limitation: Breadcrumbs free tier limits graph depth to 5 hops. W-04 (1.01 BTC) could not be traced further with the free tool. A paid tier or Chainalysis Reactor would likely resolve this.

3. TRANSACTION DETAIL

Each row below corresponds to one on-chain transaction identified during the investigation. Timestamps are in UTC. Amounts reflect minor losses to miner fees at each hop.

Date / time	From → To	Amount	Flag
02 Mar 14:32	Victim (1LogiFr) → W-01	**3.20 BTC**	Ransom
03 Mar 09:14	W-01 → W-02 (mixer)	**3.18 BTC**	Mixer
05 Mar 03:47	W-02 → W-03 (exchange)	**2.17 BTC**	Exchange
05 Mar 03:47	W-02 → W-04 (unknown)	**1.01 BTC**	Unknown
07 Mar 11:22	W-03 → KuCoin deposit	**2.17 BTC**	Fiat exit

■ The 0.02 BTC difference between W-01 (3.20) and W-02 (3.18) represents miner fees — this is normal on-chain behaviour, not a separate transaction. I initially misread this as a second output; confirmed it was fees by checking the raw transaction on Blockchair.

4. RED FLAGS AND ANALYTICAL FINDINGS

1. **Rapid chain hop within 72 hours.**
Funds moved from ransom wallet to mixer within 19 hours of receipt. This speed is consistent with automated laundering scripts that move funds before exchanges can freeze the address.

2. **Confirmed mixer use (W-02).**
W-02 (3Mix9Rp...2cKL) matched a known mixer cluster in Breadcrumbs. Mixers pool funds from multiple sources and redistribute them, deliberately breaking the transaction trail.

3. **Split output at mixer stage.**
The mixer sent 2.17 BTC to an exchange and 1.01 BTC to an unknown wallet (W-04). Splitting outputs is a common technique to reduce the traceable amount per destination.

4. **Off-hours transaction timing.**
The mixer-to-exchange hop occurred at 03:47 UTC, outside normal business hours. This may be intentional to reduce the chance of manual review at the exchange's compliance team.

5. **Exchange identified as KuCoin (DE cluster).**
The deposit address clusters to KuCoin's German entity based on Breadcrumbs' exchange tagging. This is the key finding: a legal request to KuCoin could identify the account holder.

5. CONCLUSION AND NEXT STEPS

The transaction graph traces 3.20 BTC paid as ransom through a three-hop laundering chain ending at a KuCoin exchange deposit. Approximately 68% of the original amount (2.17 BTC, ~€86,800) reached an identifiable fiat off-ramp, while 1.01 BTC remains untraced beyond the mixer output.

The combination of rapid movement, mixer use, off-hours timing, and split outputs is consistent with organised ransomware group behaviour documented in public threat intelligence sources (CISA, Europol EC3). The KuCoin deposit address is the most actionable lead for law enforcement follow-up.

File STR with SEPBLAC
Include BTC transaction hashes and the KuCoin deposit address as supporting evidence.

Submit MLA / subpoena request to KuCoin DE
Request KYC data for the account linked to bc1qEx8...9mYZ to complete attribution.

Continue tracing W-04 (1.01 BTC)
Requires paid blockchain analytics tool (Chainalysis / Elliptic). Currently unresolved.

Submit IOCs to Europol EC3
Share LockFlow wallet cluster and mixer address for cross-case correlation.

■ What I would do differently with professional tools: Chainalysis Reactor visualises the full cluster automatically and flags mixer addresses in real time. With Breadcrumbs free I had to trace each hop manually, which took longer but helped me understand the methodology better.